Quantitative Aptitude Formula Handbook

Prepo by Avneesh — Core Formulas & Shortcuts | Category: FORMULA SHEET

1. Percentages, Profit & Loss

• Standard Fractions: 1/2 = 50%, 1/3 = 33.33%, 1/4 = 25%, 1/5 = 20%, 1/6 = 16.66%, 1/7 = 14.28%, 1/8 = 12.5%
• Profit % = (Profit / CP) * 100 | Loss % = (Loss / CP) * 100
• SP = CP * (1 + Profit% / 100) or CP * (1 - Loss% / 100)
• Successive Discounts: Single equivalent discount = [d1 + d2 - (d1 * d2) / 100]%
• Faulty Balance / Weights: (True Weight / False Weight) = (100 + Gain%) / 100

2. Time & Work, Pipes & Cisterns

• Work = Efficiency * Time | If A takes X days and B takes Y days: (Together) = (X * Y) / (X + Y)
• Chain Rule: (M1 * D1 * H1) / W1 = (M2 * D2 * H2) / W2
• Alternating Work Cycles: Sum the work of one complete cycle first before remaining fraction.
• Pipes: Net filling rate = (Sum of inlet rates) - (Sum of outlet/leak rates)

3. Time, Speed & Distance, Trains & Boats

• Conversion: 1 km/h = 5/18 m/s | 1 m/s = 18/5 km/h
• Average Speed (Equal distances at speeds u and v) = 2uv / (u + v)
• Relative Speed: Same direction = S1 - S2 | Opposite directions = S1 + S2
• Crossing a platform of length P by train L: Distance = L + P
• Boats: Downstream = Speed + Stream | Upstream = Speed - Stream

4. Simple Interest & Compound Interest

• Simple Interest (SI) = (P * R * T) / 100
• Compound Amount = P * (1 + R / 100)^T
• Difference between CI and SI for 2 years = P * (R / 100)^2
• Difference between CI and SI for 3 years = P * (R / 100)^2 * (3 + R / 100)